

Global Renewable Energy in 2025

How Far Have We Come?

A data-driven overview of renewable energy's growing share in the global energy and electricity mix, spanning hydropower, wind, and solar technologies.

Source: Our World in Data · Authors: Hannah Ritchie, Max Roser, Pablo Rosado

Data: Ember (2026); Energy Institute – Statistical Review of World Energy (2025)

75% of Global GHG Emissions Still Come from Fossil Fuels

75%

of global greenhouse gas emissions from fossil fuels

5M+

premature deaths per year from fossil fuel air pollution

Since the Industrial Rev.

fossil fuels have dominated most countries' energy systems

The Path Forward

To reduce CO₂ emissions and air pollution, the world must rapidly shift toward low-carbon energy sources — nuclear and renewable technologies.

Renewable energy includes solar, wind, hydropower, geothermal, bioenergy, wave, and tidal sources.

Traditional biomass — important in lower-income settings — is typically excluded from modern renewable statistics.

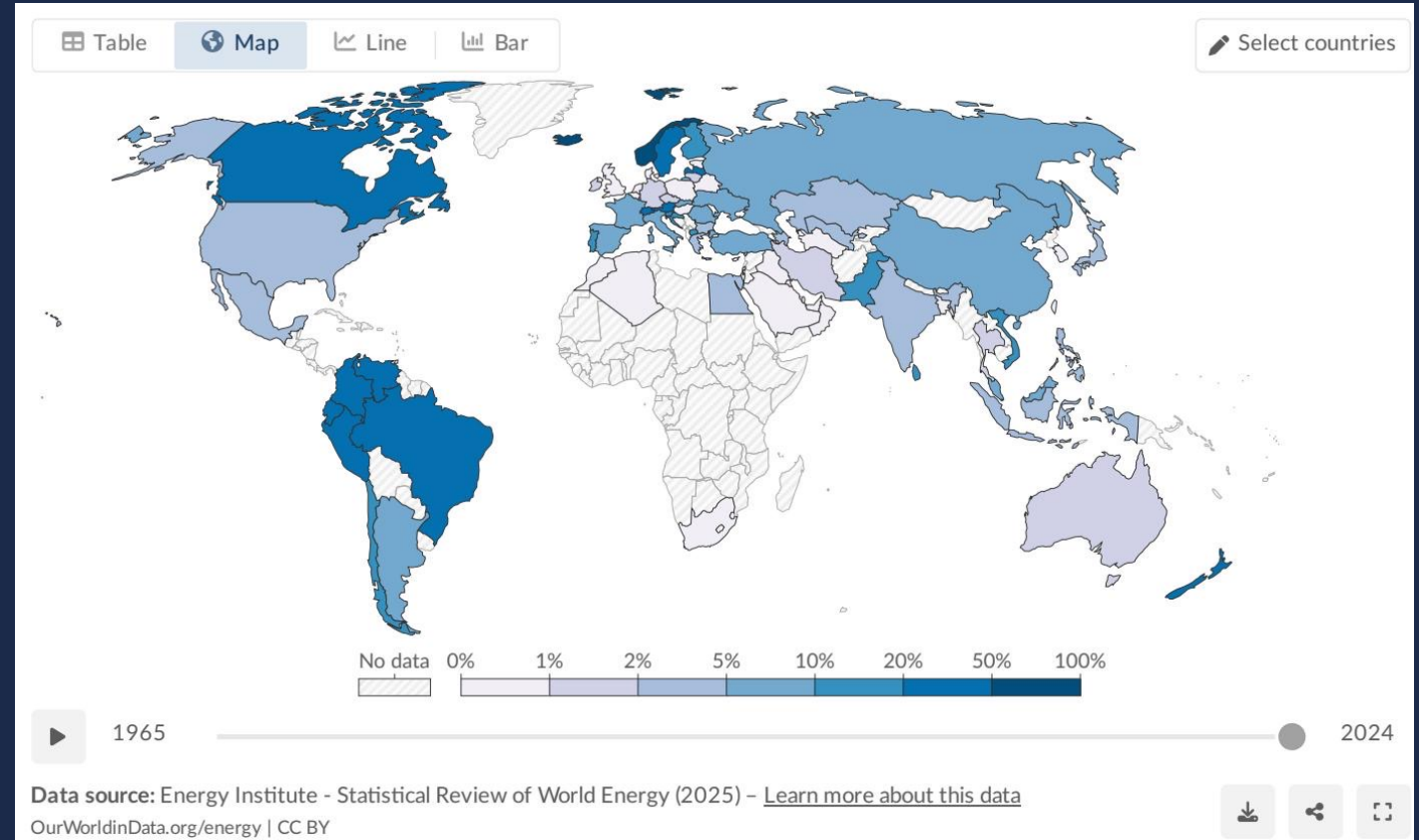
Renewables Now Supply ~14% of Global Primary Energy

Primary energy encompasses electricity, transport, and heating combined (2024, substitution method)

~14%

≈ one-seventh of global primary energy

- ▶ Transport and heating are harder to decarbonize than electricity, keeping the overall share lower
- ▶ Norway, Brazil, and New Zealand derive large shares from renewables
- ▶ Fossil-fuel-producing nations in the Middle East and parts of Asia remain below 5%



Renewable Electricity Generation Is Growing Rapidly, Led by Wind and Solar

Global renewable electricity generation by technology, 2024



~9,600 TWh

Total renewable electricity generation in 2024

Hydro: ~47%

Still the single largest renewable source

Wind + Solar: ~46%

Combined, nearly matching hydropower output

Other: ~7%

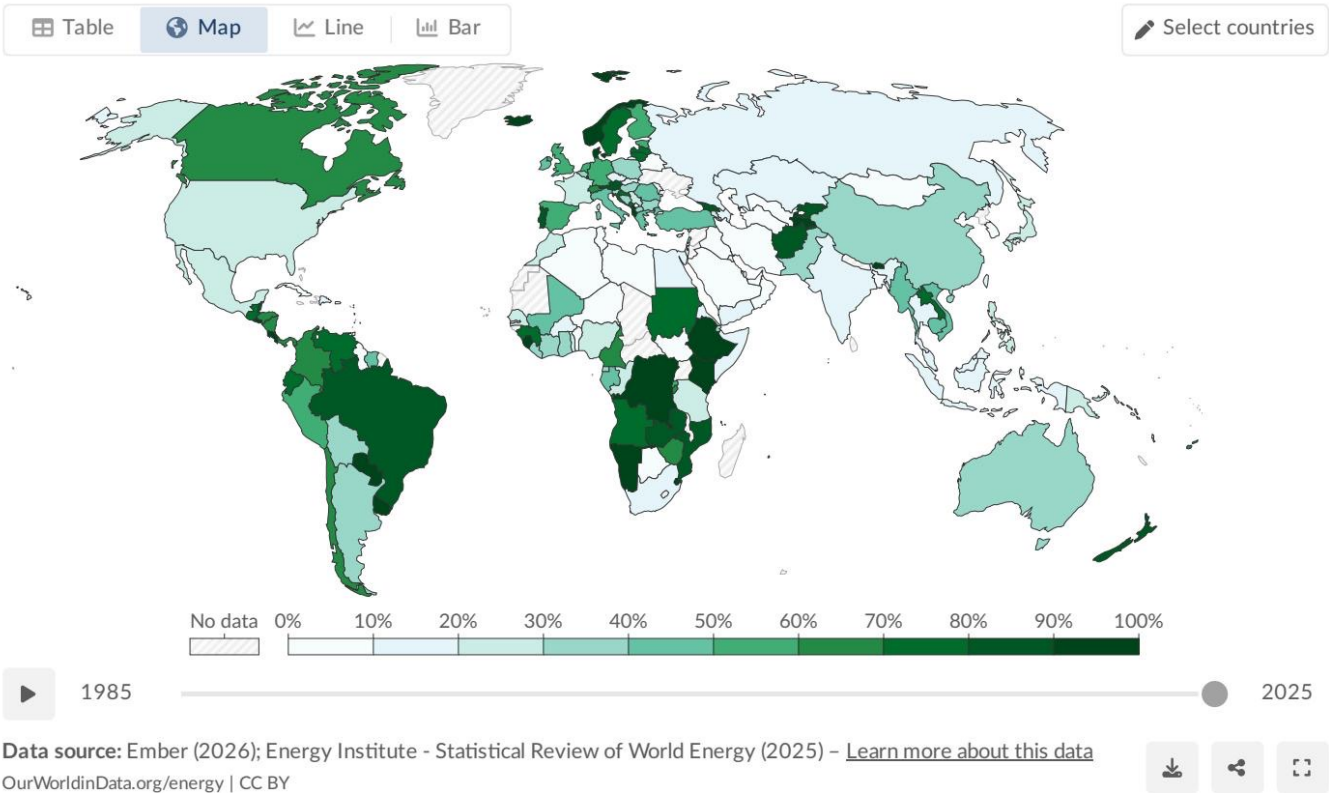
Geothermal, biomass, wave, and tidal

Almost One-Third of Global Electricity Now Comes from Renewables

Share of electricity from renewable sources by country, 2025

Share of electricity production from renewables, 2025

Renewables include solar, wind, hydropower, bioenergy, geothermal, wave, and tidal sources.



Country / Region	Renewable Share
Norway	~98%
Brazil	~85%
Canada	~65%
Germany	~55%
China	~35%
India	~25%
United States	~25%
World Average	~30%
Saudi Arabia	<2%

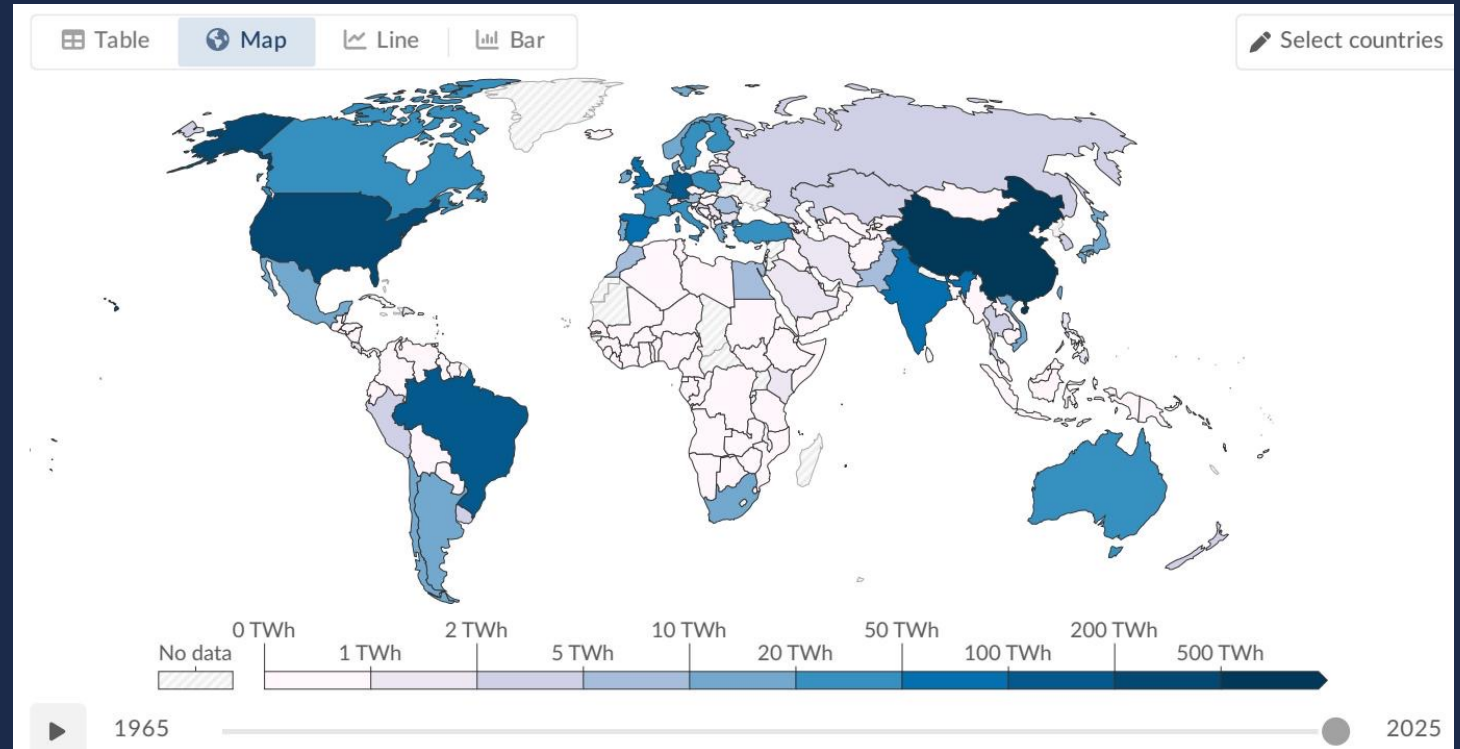
Hydropower Remains the Largest Renewable Source at ~50% of Generation

Hydropower generation by country, 2025

~4,500 TWh

Global hydropower generation (2024)

- ▶ Accounts for roughly half (~47%) of all renewable electricity
- ▶ Generating electricity at scale for over a century
- ▶ Top producers: China, Brazil, Canada
- ▶ Growth potential is limited compared to wind and solar



Wind and Solar Are the Fastest-Growing Energy Technologies in History

Wind and solar generation comparison (2024 data)

WIND ENERGY

~2,400 TWh

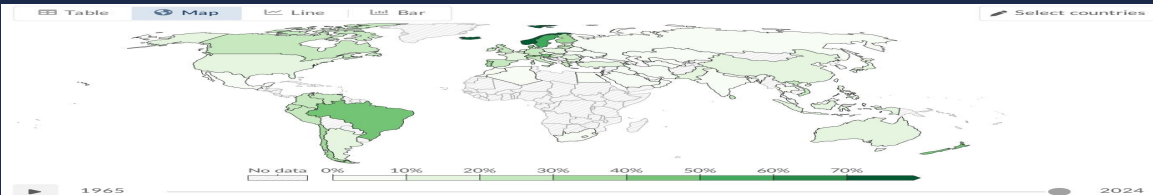
~25% of renewable electricity generation
Major contributor in Europe, China, and the US
Includes both onshore and offshore installations

SOLAR ENERGY

~2,000 TWh

~21% of renewable electricity generation
Fastest-growing source — dramatic growth since 2010
Driven by falling costs and policy support

Combined: ~4,400 TWh (~46% of renewables) — nearly matching hydropower's ~4,500 TWh output



↑ Wind power generation by country, 2025

Stark Regional Gaps: Sub-Saharan Africa Lags While Scandinavia and Latin America Lead

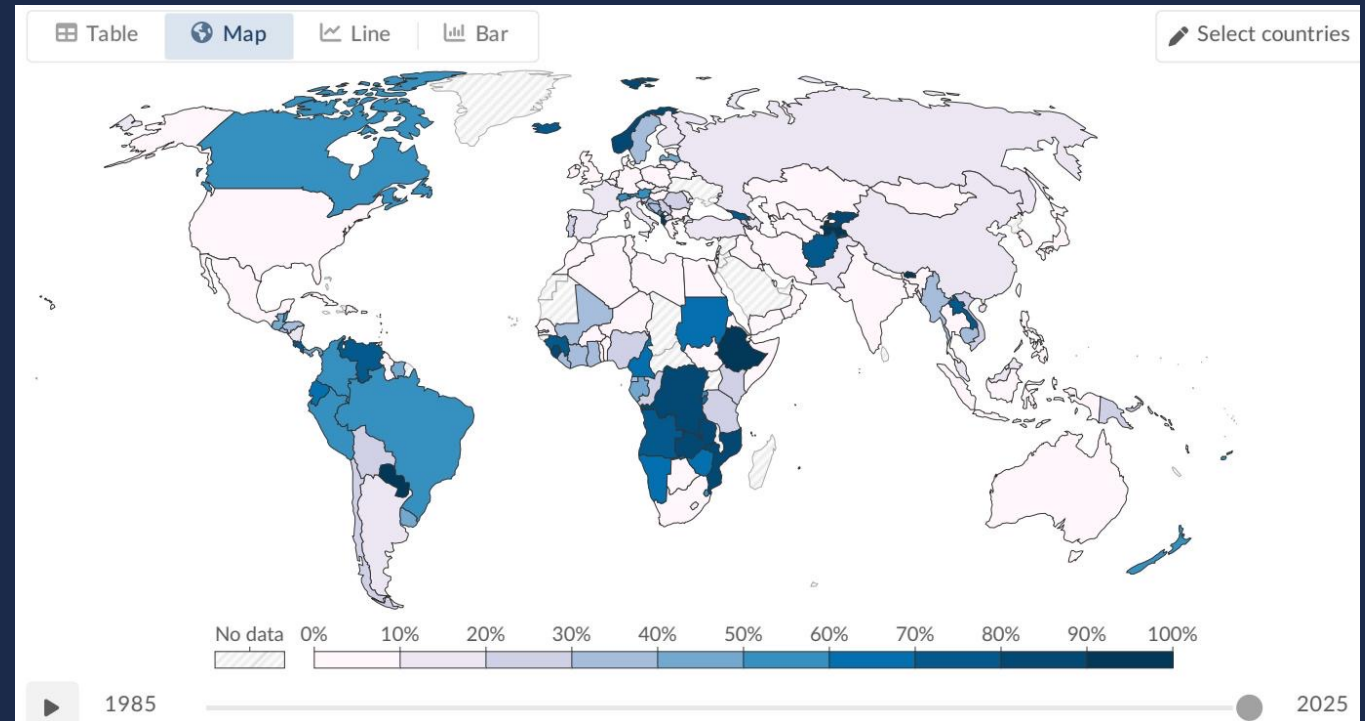
Share of electricity from hydropower by country, 2025

LEADERS (65–98% renewable electricity)

- ▶ Scandinavia & Brazil — abundant hydropower
- ▶ Denmark & Germany — heavy investment in wind and solar
- ▶ Canada — ~65% from diverse renewable portfolio

LAGGING (<10% renewable electricity)

- ▶ Sub-Saharan Africa — minimal renewable infrastructure
- ▶ Middle East — almost entirely fossil-fuel dependent
- ▶ Saudi Arabia — renewable share below 2%



Key Takeaways: Renewables Are Growing Fast but the Transition Must Accelerate

1

~14% primary energy, ~30% electricity

Renewables have reached a significant milestone — but these shares remain far from sufficient for climate targets.

2

Hydropower still dominant

Generating roughly half of all renewable electricity, but its growth potential is limited compared to wind and solar.

3

Wind & solar: fastest-growing ever

Combined, they now produce nearly as much electricity as hydropower and are accelerating.

4

Stark regional disparities persist

Scandinavia and Latin America exceed 60–90% renewable electricity; much of the Middle East and Sub-Saharan Africa remain below 10%.

5

Transport & heating lag behind

Electricity leads the transition, but overall primary energy share stays low because transport and heating remain hard to decarbonize.

The renewable transition is underway — but the pace must accelerate dramatically to meet global climate targets.